Product Design / Interaction Design / App Design

NEARBY

Bring your break with you.

INTRODUCTION

A following, height-adjustable tea table for young adults who want effortless downtime.

Keep your tea, book and phone together as you move between the balcony and sofa. Park beside your seat, adjust the height, and bring your familiar setup along for a short stay.

Following tea table · Home + short-stay living



AI concept render



Scenario Generated by Midjourney

Introduction

NEARBY is a self-following, height-adjustable tea table for young adults who want effortless downtime. Drinks, a book and a phone stay together as they move between the balcony and the sofa. A compact transport concept extends this familiar setup to short stays.

How Might We

How might we help young adults move between their favourite places to relax, without carrying a table or setting up their tea break all over again?

Through this project,
I explore how furniture can follow the rhythm of a break, adapting to where a person sits and how they relax.

Original lift-box prototype · Portable transport proposal
Modelled packing envelope · Vehicle fit testing next

Keywords

Self-following tea table / Adjustable height / Portable setup

Tools

Figma / Blender / STM32H743 / OV2640

Background

What is NEARBY?

A self-following, height-adjustable tea table that keeps drinks and everyday essentials close as young adults move between their favourite places to relax.

Read on the balcony, then unwind by the sofa. Your tea setup comes along, without carrying the table or gathering everything again.

Project Context: Home + Short-Stay Living


Tea by the window


Flexible living spaces

HOME
Living room
+ enclosed balcony

AWAY
Short-stay
accommodation

BREAK
Tea · reading
· relaxation

FOLLOW + LIFT
Move with you · fit your seat

Young Adults
Effortless downtime, wherever you sit

Main Characteristics

Move the break with you
Keep a cup, book and phone together as you switch between your indoor resting places.

Fit different ways of sitting
Use a lower surface beside the sofa or raise it for a chair. Adjust the height after parking.

Pack for the next stay
Lower and lock the table for travel. A pull handle and side grips support manual towing and two-hand loading.

Current Situation

A relaxing break can involve small but repeated tasks: carrying drinks, moving a table and resetting a place to sit. The project explores how furniture could move with the user.

Carrying	Moving	Resetting
/ cups + phone + book / occupied hands / repeated trips	/ table left behind / changing rooms / clearing a route	/ new sitting position / mismatched height / interrupted leisure

Desk Research / Tabletop Heights


Coffee table
45 cm
VITTSJÖ [1]


Sofa side table
60 cm
ASPERÖD [2]

Seated desk
75 cm
TROTTEN [3]

Photos & dimensions: IKEA [1–3]. Height above floor.

Design aim: let the tea break move with the person. Combine controlled following and parked-height adjustment, with compact manual transport for short stays.



Basis: creator's brief. Context photos: Pexels / Gary Barnes, Max Vakhtbovych. Cover: AI concept simulation. Benefits not measured.

Scenario Research

Home and short-stay leisure · Scenario study from the brief
Questions for future observation, not completed fieldwork.

Pain Point Themes

Human–Object Interaction: Carrying a complete tea break



Several items to carry

Hands already occupied



A book and cup share the seat

Essentials need a nearby surface



Different ways to sit and rest

Surface height needs to adapt



Different seat heights

Setup starts over again

Illustrative context photographs · carrying, changing places and settling in are design hypotheses.

Human–Space Interaction: Switching the place to relax



1. Indoor movement



2. Balcony threshold



3. Leisure setup

Relax across rooms.

Pack, manually tow and load for the next stay.

New study questions.

Human–Furniture Interaction: What should change?

1. Move with me

2. Fit my seat

3. Let me pause

Could following and height adjustment reduce carrying and setup?

Compare with a manual table.

Context photos: Pexels; illustrative scenarios, not participant evidence. Source URLs recorded in image layer names.

Scenario Analysis & Insights

User Groups

Homebody
Young adult · tea and reading
Moves between favourite seats

Short-stay guest
Young adult · short-stay travel
Wants a towable, stowable setup

Shared-home user
Young adult · shared living spaces
Wants clear, controllable motion

Recruitment targets, not completed interviews.

Problem Redefinition

What interrupts an easy indoor break?
Three questions derived from the brief.

Carrying the whole setup
Can cup, book and phone move together without repeated trips?

Changing resting places
Can the table follow when comfort or light sends me elsewhere?

Fitting the next seat
Can a parked surface adapt to a sofa, chair or relaxed posture?

Design opportunity: less carrying indoors + compact transport between stays.

Design Lens: Leisure Interruptions

- Observe tea setup, room changes, packing and car loading.
- Compare a manual side table and luggage; measure effort, stability and handling.

POSSIBLE EXPERIENCES

Ready · Carrying · Interrupted · Relaxed

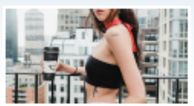
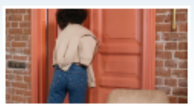
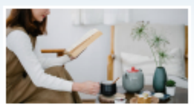
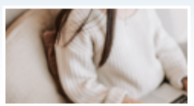
Setup tasks → Perceived effort → Break interrupted

Explore when a change of place interrupts leisure and what movement people would accept.
The relationships and feelings shown are hypotheses for a future study.

Basis: brief-led scenario analysis · validate through observation and comparative trials.

User Journey & Effort Map

BASELINE SCENARIO / Balcony tea → living-room rest with a fixed table · hypothesis

	1 PREPARE	2 CARRY	3 SETTLE	4 RELOCATE	5 RETURN	6 RESET	7 ADJUST	8 RELAX
Possible questions	*What do I need for a break?*	*Can I carry everything?*	*Where shall I sit first?*	*Shall I move to the sofa?*	*Did I leave my phone?*	*How do I move the table too?*	*Is this height comfortable?*	*Can I relax again now?*
Practical demands	Gather the tea setup	Balance several items	Choose a resting place	Change the setting	Retrieve a forgotten item	Carry and reposition	Fit the new seat	Resume the break
Emotional experience	😐	☹	😐	🙂	😐	☹	😐	🙂
Actions	 Tea preparation context	 Tray + small essentials	 Balcony break context	 Room-to-room movement	 Doorway / return route	 Next seated tea setup	 Seated reach context	 Sofa reading context

Design Concept

Solution Integration

Your tea break, on the move.
Following + adjustable furniture for downtime at home and on a short stay.

Young adult / direct controls
FOLLOW · STAY · HEIGHT

Personal app
Save places · Set height · Check charge

Tea table + indoor map
Tea setup · Parking places · Visible state

Design Goals

01 **Bring the whole break along.**
Keep a cup, tea flask, book and phone on one mobile surface.

02 **Move between indoor leisure spots.**
Start FOLLOW, change rooms and stop beside the next seat.

03 **Fit the seat. Pack for the next stay.**
Adjust after parking; lower and lock for manual transport.



System Integration

The table and app share following, height and packing states.
Stop before packing, then use the pull handle and side grips.

Technology

Prototype: wheeled cabinet + lift.
Proposed: locks, grips + towing.

Technical Research

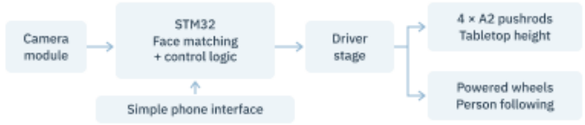
Embedded Vision + Motion Control

On-device vision [1]
Local face matching links the device to a selected user. Identity matching and continuous person tracking are separate engineering tasks.

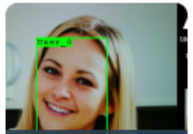
Four-point electric lift [2]
Four A2 motor-driven electric pushrods raise and lower the tabletop. Coordinated travel, stopping and load stability need verification.

Following + phone commands [3, 4]
The powered base is designed to follow the selected user. The proposed app adds start, stop and parked-height controls; integration testing is next.

Basic Structure / Functional Architecture



Technology Applications / Reference Examples



01 Recognise
On-device face matching
ST application [1]



02 Lift
Electric linear motion
Actuator reference [2]

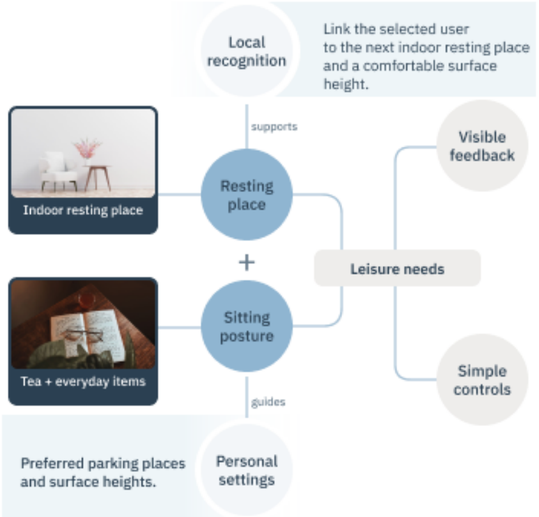


03 Move
Motor-driven wheelbase
Drive reference [3]

[1] ST / FP-AI-FACEREC1 [2] Bansbach / Electric actuators
[3] Pololu / 3pi+ 2040 [4] LINAK / Desk Control App

Concept Development

How Does NEARBY Help?



Advantages to Validate

Continuity
Bring the break with you

Control
Start and pause at any time

Less carrying
Keep essentials together

Comfort
Fit the next sitting position

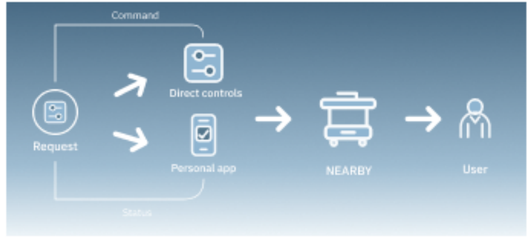
Simple setup
Save a place and height

Core Topic

A tea break that moves with you.
Follow indoors, park and adjust;
pack for manual travel between stays.

REQUEST
- FOLLOW begins a session.
- STAY stops; HEIGHT adjusts.

FEEDBACK
- Following / Paused / Ready
- Visible feedback for the next action.



1 Choose FOLLOW, STAY or HEIGHT.
2 Park before reaching or changing height.
3 Stop before packing; clear the tray, lower and lock. Tow manually with following disabled.

Essential Information

Next-action information
Show only what matters at this step.

- Cup · tea flask
- Book · phone

- Indoor places
- Clear routes

- Following · parked
- Height + battery

System Design

System Map

External connections

Young adult

Room setup + presets

Service + repair

Local control

Camera + local matching.
Lift and follow commands
Visible stopped states.

Local control STM32 + camera

Local settings & recognition

Personal setup

Choose tea essentials.
Save places and heights.
Pack for the next stay.

NEARBY

Personal app

Tea + leisure items

Direct controls

Device Touchpoints

Start or pause following.
Adjust while stationary.
Check charge and status.

User app

Direct control

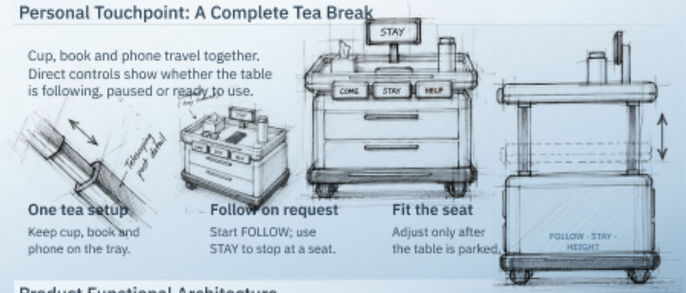
Tea table

Follow indoors. Park before adjusting height.
Pack and lock before manual transport.

Touchpoint Definition

Personal Touchpoint: A Complete Tea Break

Cup, book and phone travel together.
Direct controls show whether the table is following, paused or ready to use.



One tea setup
Keep cup, book and phone on the tray.

Follow on request
Start FOLLOW; use STAY to stop at a seat.

Fit the seat
Adjust only after the table is parked.

Product Functional Architecture

- Keep leisure items together.
- Follow and park between seats.
- Adjust while stationary.
- Lower, lock and manually transport.

- FOLLOW: start a following session.
- STAY: stop; HEIGHT: set the surface.

- Save places and preferred height.
- Guide packing and manual checks.

Device Touchpoints

A Tea table
Mobile storage · adjustable tray
Visible movement and height state

B Direct controls
FOLLOW / STAY / HEIGHT
One-touch movement and adjustment

C Personal app
Places, presets and packing checks
Review status before transport

Essential Functions
Mobile tea storage with direct following and height controls.

Following + Lift
Carry the break between seats, then fit the surface height.

Extended Functions
Personal places, height presets and a guided packing checklist.

Touchpoints + Environment

Space Management

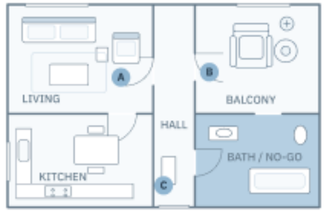
Home + short-stay review

- Identify living-room and balcony seats.
- Check doorway clearance and thresholds.
- Review the current accommodation.

Adjustable settings

- Save a few indoor parking places.
- Mark wet areas and door thresholds.
- Confirm a clear route for following.
- Pause if the target or route is unclear.

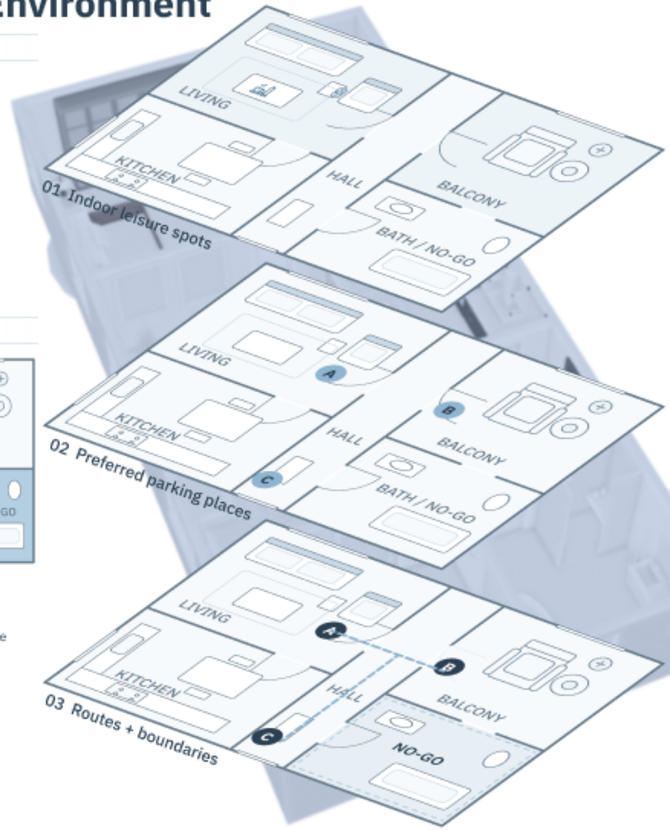
Product Layout Principle



A / Living-room sofa
Park beside the sofa with tea nearby.
Adjust the surface for a relaxed seat and keep the walking route clear.

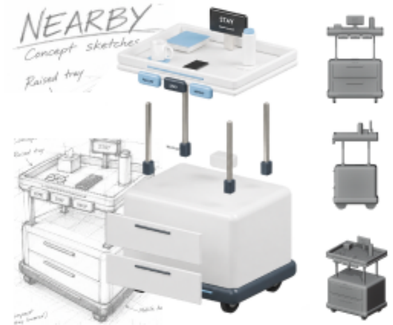
B / Balcony chair
Use an enclosed balcony on the same level.
Review the doorway threshold before enabling a following route.

C / Charging place
Return to a clear charging spot.
Review the room layout at each new short-stay accommodation.



Device design

Proposed component architecture



Leisure mode → packed travel box

New: pull handle, side grips and transport locks.
Vehicle fit and handling strength require tests.



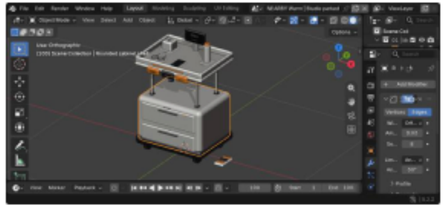
01 Lower + lock



02 Manual tow



03 Car / AI concept



Blender modelling view · concept renders



Scenario Generated by Midjourney · parked tea break; proposed controls and appearance

App Design

Personal leisure controls



Follow & adjust

Start following.
Park by the next seat.
Set the surface height.



Home





Save a favourite place
and tabletop height.

Room presets & tea setup

Save places.
Keep your
tea setup
ready.



Direct control

Start or pause,
check charge
and adjust
a parked table.



Pack Mode / Manual Transport



LEISURE → TRAVEL

Clear → lower → lock
Turn following off.
Confirm checks yourself.

TOW & LOAD

Extend the pull handle.
Carry with both side grips.
Secure inside the car.

Proposed mode
Locks and car fit need testing.

Visual Guidelines

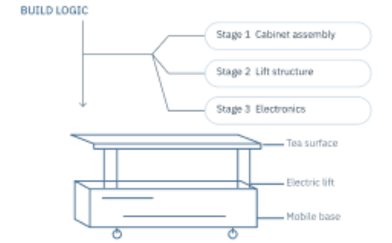
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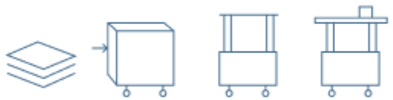
IBM Plex Sans

Colour

Prototype | Following tea table

From a lift-box demonstrator to a leisure tea table: cabinet assembly, electric lift and control integration.





Panel kit — Cabinet → lift → integrated prototype

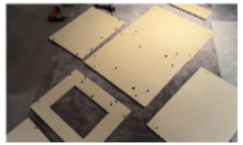


Base assembly / Fix the caster brackets and wheels.



Control preparation / Wiring notes and circuit connection.

Mechanical assembly / steps 01–06



01 Prepare the panels



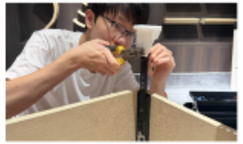
02 Fasten the cabinet



03 Fit the drawer rails



04 Mount lift hardware



05 Align the guide rails



06 Tighten the fasteners

Wiring preparation / steps 07–12



07 Solder the control board



08 Prepare actuator connections



09 Wire the control board



10 Connect the cable set



11 Route cables inside the cabinet



12 Fit the display and connectio

Original lift-box demonstrator

STM32H743 + OV2640 · fixed ROI
PC-tested core; board testing pending
Face matching supports pairing; tracking is separate.





Source code on GitHub ↗

A break that follows



Reflection + Next Test

This project reframes convenience as keeping a break together while moving between spaces. The next step is to compare NEARBY with carrying a conventional table, testing following, stopping and drink stability. The travel extension also needs checks for lifting effort, locks and vehicle fit.

AI concept simulation · movement behaviour and link stability not yet validated